Muhammad Ghazi Randhawa

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Education

Swarthmore College Swarthmore, PA
B.A. in Economics & Computer Science *2018 – 2022*

- **Economics Coursework**: Antitrust Economics and Regulation, Environmental Economics, Law and Economics, Financial Accounting, Global Environmental Governance
- **Mathematics**: Probability, Mathematical Statistics, Statistical Methods, Discrete Mathematics, Calculus, Linear Algebra
- **Computer Science**: Operating Systems, Algorithmic Game Theory, Machine Learning, GIS, Data Structures, Programming

Teaching and Work Experiences

Associate Data Scientist March 2025 - Current
Data Pilot

Empirical Research Fellow July 2023 - July 2024
Stanford Law School

Predoctoral Research Fellow August 2022 - June 2023
Stanford Institute of Economic Policy Research

Skills

Programming Languages: C, C++, Python, SQL, Bash, CUDA(familiar), MPI(familiar), OpenMP(Basic Experience), OpenCL(Basic Experience), R, Stata, and GIS

Frameworks/Libraries: Python (scikit-SkLearn, OS, Pandas, Numpy, Matplotlib, Seaborn, YAML, Selenium, and other libraries), standard C and C++ libraries, OS and Networking fundamentals

Machine Learning: PyTorch, Tensorboard, CV, Langchain, Prompt Engineering

Data Engineering: Airflow, Dagster

Cloud Frameworks: Azure devops, GCP

Developer Tools: Jupyter Notebook, Version Control(Git), Latex, Virtual Box, Selenium

Research Experiences and Projects

Pollution misreporting in Clean Water Act Stanford, CA
SLS-,SIEPR-fellow *August 2022 – Present*

- Used Python, R, and Postgres to develop a forensic accounting data science tool to determine levels of incidence in fraud by third-party water testing facilities. It uses EPA's data for 26 states that have implemented electronic report submissions. The project aims to develop similarity distributions using Levenstien-distance for report-submitter for four different groups of reported values: First-party reporters, Third-Party reporters for same Permit, Third-Party reporters across different Permits, and Placebo pairs.
- For these four groups, this similarity score is used to calculate the duplication rate for each permit and each third-party. We found that third party provider labs tend to have lower fraud ratings, but certain third parties lab commit suspicious copying behavior.
- The Policy aim of the project is to help develop data-centric environmental regulatory methods that minimize the risk of fraud

Effects of Mandatory reporting rule changes on OTC companies Stanford, CA
SLS-,SIEPR-fellow *August 2022 – Present*

- Developed a python-Selenium script to webscrape 5000 disclosure documents. Developed pytesseract script to properly scan these complex documents
- Used Regex and text-centric data techniques to develop datasets about these companies' Shareholder and Equity information, Third-Party Advisors, Transfer Agents & other relevent information for economics of securities regulations

- Worked on investigating some of these entities and professionals to determine if dubious companies tend to work with dubious advisors

Automated Metrics for Surgical fine-tuning for Domain Shifts Autumn 2022

Metalearning

- Implemented and Compared fine-tuning methods (Full-Finetuning, Linear Probing, Cross-Validation, Auto-RGN (Relative-Gradient-Norm), Auto-SNR (Signal-to-Noise Ratio), & Auto-RGNxSNR) in Transfer-learning for adaptation to distribution shifts on 3 benchmark distribution-shift datasets (celebA, waterbirds, Camelyon17) using Resnet-50.
- Verified results in prior literature that indicate that Auto-RGN & Auto-SNR methods of surgical fine-tuning can consistently perform better or no worse than Cross-validation Finetuning of layers.
- Investigated whether Auto-RGN and Auto-SNR can achieve better results by being used together.
- My results found that for the Waterbirds, Auto-RGNxSNR performed better than Auto-RGN & SNR methods. For Camelyon17, I found that Auto-RGNxSNR performed better than Auto-SNR but worse than Auto-RGN. CelebA dataset showed that Auto-RGNxSNR performed worse than the other metrics.

Speed-up Analysis of Three Sorting Algorithms Fall 2021

Parallel and Distributed Programing

- Implemented sequential and parallel versions of three (Odd-Even, RadixSort, Bitonic) sorting algorithms in MPI and C, and ran experiments for sizes upto 10 million elements per process $\frac{N}{P}$. Parallel tests were conducted on $4, 16, 32, 64, 128$ and 256 processes.
- For sequential experiments , Radix and Bitonic sort had better runtime in the number of elements.
- For parallel experiments, user runtime increased with $\frac{N}{P}$ for odd-even and bitonic sort for largest values of P and remained relatively constant for smaller number of P. As per intuition, the system runtime for odd-even and bitonic sort algorithms increased as we increased with the number of processes and number of elements per process. Interestingly for Radix, the user runtime decreased with increased number of P and $\frac{N}{P}$.

Machine Learning Models to predict Asthma rates Spring 2021

Machine Learning

- Project trained four machine learning models (Random Forests, Support Vector Machines, Logistic Regression, and ID3 decisions) using SKlearn Packages to be able to classify at-risk communities for Asthma.
- Created a novel dataset for 324 counties across 6 US states was created of environmental, health, and demographic variables with two different outcome prediction scheme(four labels vs 2 labels).
- After cross-validating and hyper-tuning the models, we looked at the feature importance methods 3 of these 4 models (SGD: weights, Decision Trees: Decrease in Impurity, Random Forests: Permutation Feature Importance & decrease in impurity in member decision trees, SVM: Feature importance).
- We found that Random forests and Decision Trees consistently placed high weight on population of certain racial minorities in classifying a county as high risk for asthma relative to other features.

Department of Engineering, Swarthmore College Swarthmore

Research assistantship with Professor Arthur Macgarity *January 2019 – March 2020*

- **Effect of rainwater runoff on edible plants**: From January 2019 to May 2019, worked on a project to determine whether plants grown on rainwater runoff are fit for human consumption. Rehabilitated Swarthmore's wetland research station and operated equipment to collect rainwater runoff samples. Grew 20 amarinth plants each in 4 different groups(runoff, distilled water, tap water, & distilled water with some heavy metals) and eventually harvested them. Dissolved edible biomass and performed spectrophotometry to determine absorption of heavy metals.Final results found that runoff fed plants absorbed metals into biomass at a much higher rate than the control groups.

- **Lab assistantship**: From August 2019 till April 2020, worked as a lab assistant. I worked on setting up and calibrating environmental and solar lab equipment in the new engineering building.

Analysis of Pollution in Ravi River Spring 2017

Environmental Management

- Took Water samples across 6 different points from River Ravi, conducted water quality tests, and assessed the results against identified uses
- Found significant uptick in pollution in the data points near Lahore
- Conducted survey of communities near the sample collection point to assess the sources and uses of water. Also Asked questions about perceived and actual health effects of the pollution on these communities
- Found widespread use of water that was unfit for intended uses of irrigation, drinking, and other purposes. Found awareness of environmental and health effects on communities and individuals surveyed, and complaints about government apathy to the grave situation.

Community work

Member of Advisory Council for Computer Science	Swarthmore, January 2020 - May 2022
Board Member of South Asian students club	Swarthmore, March 2019 - May 2022